SmileMate

Your Daily Dose of Happiness

MINI PROJECT DOCUMENTATION

A simple and interactive website that encourages happiness and positivity through motivational messages, cheerful emojis, colours, and voice support.

Project Category	Frontend Web Development
Technologies	HTML, CSS, JavaScript
Application Type	Responsive, browser-based website
Backend / Database	Not required for the current version

“Keep Smiling and Spread Happiness!”

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This document describes the current SmileMate version. It is designed as a small beginner-level frontend project.

1. Abstract

SmileMate – Your Daily Dose of Happiness is a simple, interactive, and user-friendly website developed using HTML, CSS, and JavaScript. The main purpose of the project is to spread happiness, positivity, and motivation through a colourful digital experience. Users can generate random motivational messages, change background colours, listen to messages, save favourite quotes during the current session, and switch between Telugu and English. The website also includes a smile counter, message-sharing option, and dark mode. SmileMate does not require a backend or database in its current version. It demonstrates how basic frontend technologies and browser features can be combined to create an engaging web application.

2. Introduction

A website can do more than provide information; it can also create a positive and enjoyable experience. SmileMate is designed around this idea. When users open the website, they see a friendly welcome screen with a cheerful emoji and a positive message. They can interact with different buttons to receive another message, change the page colours, listen to the message, or save a quote they like. The interface uses soft pastel colours, rounded cards, and simple controls so that users can understand it easily.

3. Problem Statement

People may experience stress, boredom, or a lack of motivation during their daily routines. Finding a quick and simple source of positive encouragement should not require a complicated application or account registration. The problem addressed by SmileMate is the absence of a lightweight, easy-to-use platform that presents short positive messages in an enjoyable way.

4. Proposed Solution

SmileMate provides a browser-based happiness website where users can view random motivational messages and cheerful emojis with one click. It adds simple interactive options such as background colour changes, copy, voice, sharing, favourites, language selection, and dark mode. The solution is intentionally small and does not collect personal information or require user login.

5. Objectives and Scope

Objectives

- Spread happiness and positivity through short motivational messages.
- Provide an attractive and easy-to-use interface.
- Allow users to view messages in English and Telugu.
- Demonstrate JavaScript event handling and browser APIs.
- Build a responsive beginner-friendly website using only frontend technologies.

Scope

The current project runs in a modern web browser. It provides a fixed collection of messages, random message selection, theme changes, voice output where supported, copying and sharing where browser permissions and security requirements allow, and a temporary favourites list. It does not include user accounts, cloud storage, an administration panel, or a server-side database.

6. Requirements

Hardware Requirements

Component	Minimum / Recommended Requirement
Computer or laptop	Any basic desktop or laptop capable of running a modern browser.
Processor	Basic dual-core processor or better.
RAM	2 GB minimum; 4 GB or more recommended.
Storage	A few MB of free space for the project files.
Display	Any standard display; responsive layout also supports mobile screens.

Software Requirements

Software	Purpose
Operating system	Windows, macOS, or Linux.
Code editor	Visual Studio Code or any text editor.
Web browser	Google Chrome, Microsoft Edge, Firefox, or another modern browser.
Live Server (optional)	Conveniently runs the HTML file on a local development server.

Other Requirements

- An internet connection is not required for the core features once the file is available.
- Clipboard and sharing features may require localhost or HTTPS, depending on the browser.
- Voice output depends on the browser and the speech voices installed on the device.

7. Technologies Used

Technology	Use in SmileMate
HTML5	Creates the page structure, headings, buttons, message area, and sections.
CSS3	Controls colours, layout, typography, responsive design, hover effects, and dark mode.
JavaScript	Selects random messages and handles button actions, favourites, counter, and language selection.
Browser APIs	Clipboard API, Web Share API, and Speech Synthesis API provide copy, share, and voice features where supported.

8. Functional Requirements

- FR-01: The website shall display the SmileMate title and a welcome message.
- FR-02: The New Smile button shall select and display a random motivational message and its emoji.
- FR-03: The Change Color button shall change the page background to a selected gradient.
- FR-04: The Copy Message button shall copy the currently displayed message when clipboard access is available.
- FR-05: The Listen button shall read the current message using browser speech synthesis when supported.
- FR-06: The Share button shall share the current message when the Web Share API is available; otherwise, it shall use the copy fallback.

- FR-07: Users shall be able to choose English or Telugu message text.
- FR-08: Users shall be able to add a message to favourites and remove it from the list.
- FR-09: The smile counter shall increase when a new message is generated.
- FR-10: Users shall be able to switch between dark and light appearance.

9. Non-Functional Requirements

- Usability: Controls and messages should be clear and easy to understand.
- Performance: The page and message updates should respond quickly.
- Responsiveness: The layout should adapt to desktop and mobile screen sizes.
- Compatibility: The core website should work in modern web browsers.
- Reliability: Buttons should perform their intended actions without breaking the page.
- Accessibility: Text should be readable and controls should have clear labels.
- Privacy: The current version does not ask users to create accounts or submit personal details.

10. System Workflow

Step 1: User opens the SmileMate website.
Step 2: The website displays the welcome message, emoji, and available controls.
Step 3: User selects an action such as New Smile, Change Color, Listen, or Add to Favourites.
Step 4: JavaScript processes the selected action.
Step 5: The page updates with the new message, colour, voice output, or favourites list.

Workflow Summary

Input	Processing	Output
Click New Smile	JavaScript selects a random message	New quote and emoji
Click Change Color	A gradient is selected randomly	Updated page background
Choose language	Selected language property is read	English or Telugu message
Click Listen	Speech Synthesis API is called	Spoken message, if supported
Add Favourite	Message is added to the favourites array	Favourite quote appears in the list

11. Project Modules

1. Message Generator

Selects a random message from the predefined message collection and displays its matching emoji.

2. Background Colour Changer

Changes the background using predefined pastel gradient combinations.

3. Language Selection

Displays the available message in English or Telugu based on the user's selection.

4. Copy and Share

Copies the displayed message to the clipboard or shares it through supported browser/device sharing options.

5. Voice Support

Uses the browser's speech synthesis feature to read the displayed message aloud. Telugu voice availability depends on the device and browser.

6. Favourite Messages

Allows users to add a displayed quote to a temporary list and remove it when no longer needed.

7. Smile Counter and Theme

Tracks how many new messages have been generated during the current page session and toggles dark/light appearance.

12. Implementation Details

The project is implemented in a single HTML file. The HTML section defines the page elements. CSS styles the page and makes it responsive. JavaScript stores the message collection in an array and uses functions to update the page when a user clicks a button. The current version uses in-memory JavaScript variables for the smile count and favourites. Therefore, those values reset when the page is refreshed. No backend server or database is needed.

Suggested Project Folder Structure

```
SmileMate/
  ■■■ index.html
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13. Testing

Test ID	Test Case	Expected Result
TC-01	Open index.html in a browser	SmileMate home page is displayed.
TC-02	Click New Smile	A message and matching emoji are updated.
TC-03	Click Change Color	The background gradient changes.
TC-04	Select Telugu	Telugu message text is displayed.
TC-05	Click Copy Message	Message is copied when browser access is allowed.
TC-06	Click Listen	The browser reads the message if voice support exists.
TC-07	Add and remove a favourite	The favourite list updates correctly.
TC-08	Toggle dark mode	Dark/light appearance changes.
TC-09	Open on a mobile-sized screen	Content adjusts to the screen width.

14. Advantages and Limitations

Advantages

- Simple to build, run, and understand.
- No backend, database, or login is required.
- Provides multiple interactive features in one small project.

- Can be used on desktop and mobile browsers.
- Useful for practising HTML, CSS, JavaScript, and browser APIs.

Limitations

- Messages are predefined; the current version does not generate AI-based quotes.
- Favourites and the smile counter are not saved permanently.
- Voice quality and Telugu voice availability depend on the user's device.
- Copy and share features may be restricted when the page is opened directly as a local file.

15. Future Enhancements

- Use localStorage to save favourites and the smile counter after page refresh.
- Add more motivational messages and categories such as study, friendship, and confidence.
- Add a daily message that changes once per day.
- Improve voice settings and provide a voice-selection option where available.
- Add a more accessible interface with keyboard support and stronger contrast options.
- Add optional AI-generated messages in a future version.

16. Expected Outcome

The expected outcome is a working, responsive happiness website where users can generate positive messages, enjoy cheerful emojis, change the page background, listen to messages, save favourites temporarily, and use English or Telugu. The project also demonstrates practical frontend development skills.

17. Conclusion

SmileMate shows how a small web application can provide a pleasant and interactive user experience using basic frontend technologies. It combines motivational content with simple browser features in a colourful and accessible interface. The project is suitable for beginners and can be improved later with permanent storage, more content, and additional language and voice options.

Keep Smiling and Spread Happiness! ■